

Recognizing the Use of Steganography in Forensic Evidence (4e)

Digital Forensics, Investigation, and Response, Fourth Edition - Lab 02

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Time on Task:

2 hours, 28 minutes

Progress:

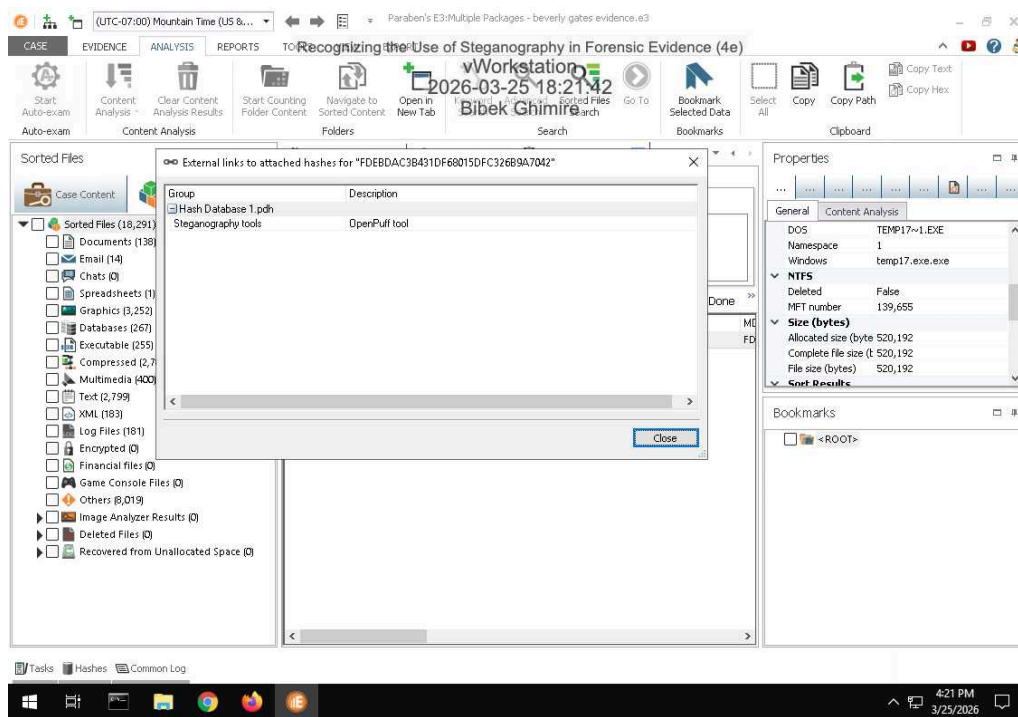
100%

Report Generated: Thursday, June 25, 2026 at 4:40 PM

Section 1: Hands-On Demonstration

Part 1: Detect Steganography Software on a Drive Image

14. Make a screen capture showing the search result and its description.

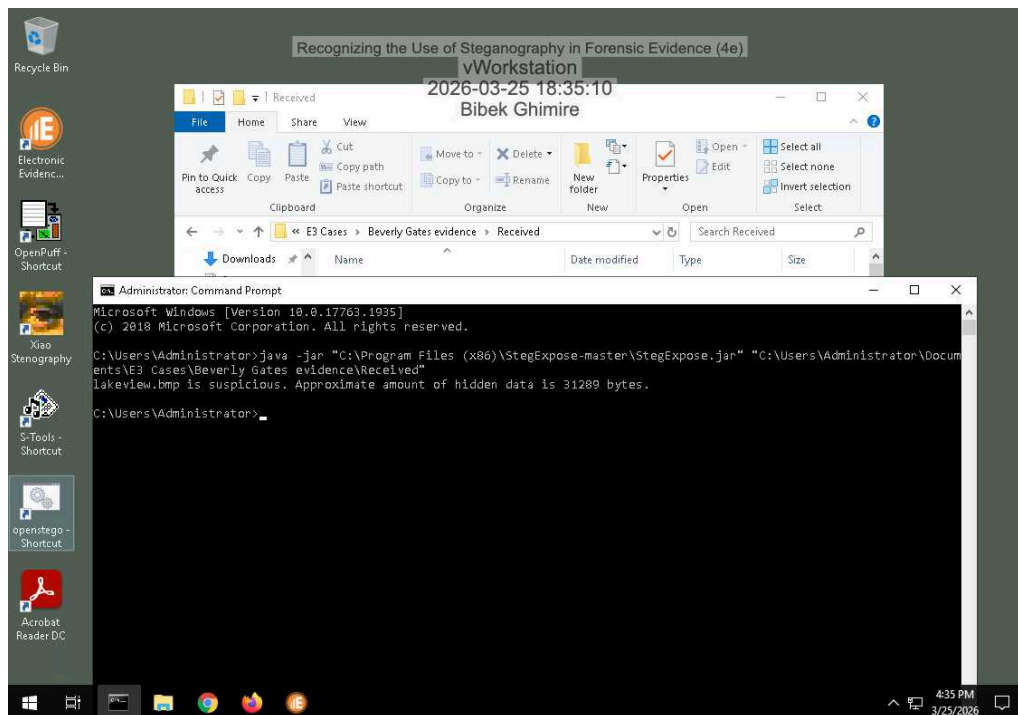


Part 2: Detect Hidden Data in Image Files

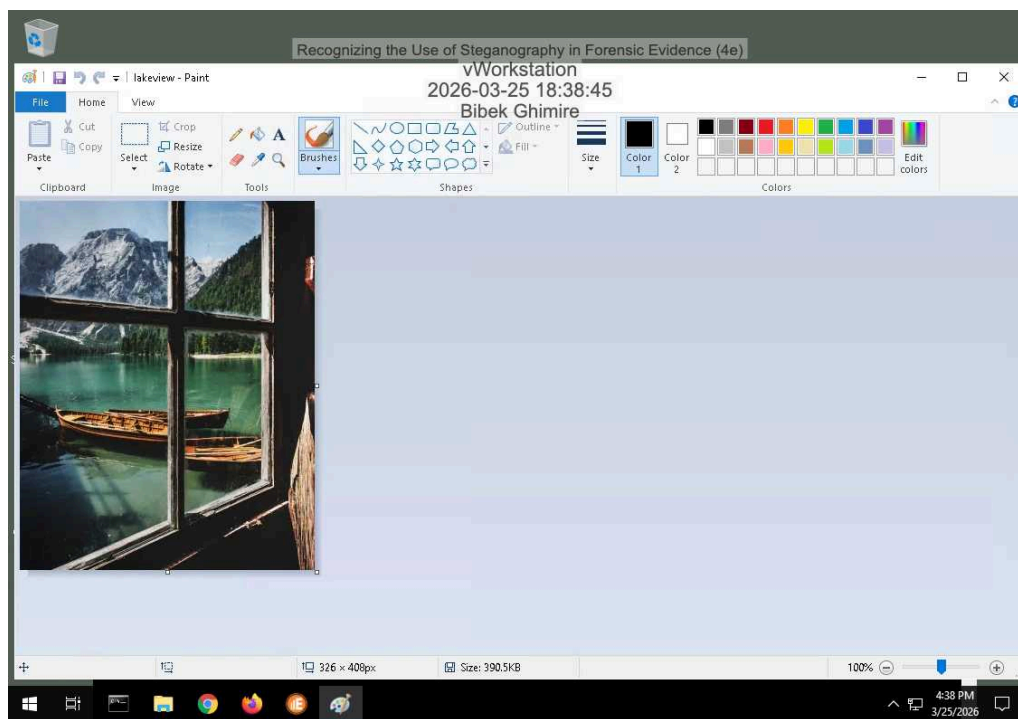
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10. Make a screen capture showing the **StegExpose** results.



13. Make a screen capture showing the suspicious file in Microsoft Paint.

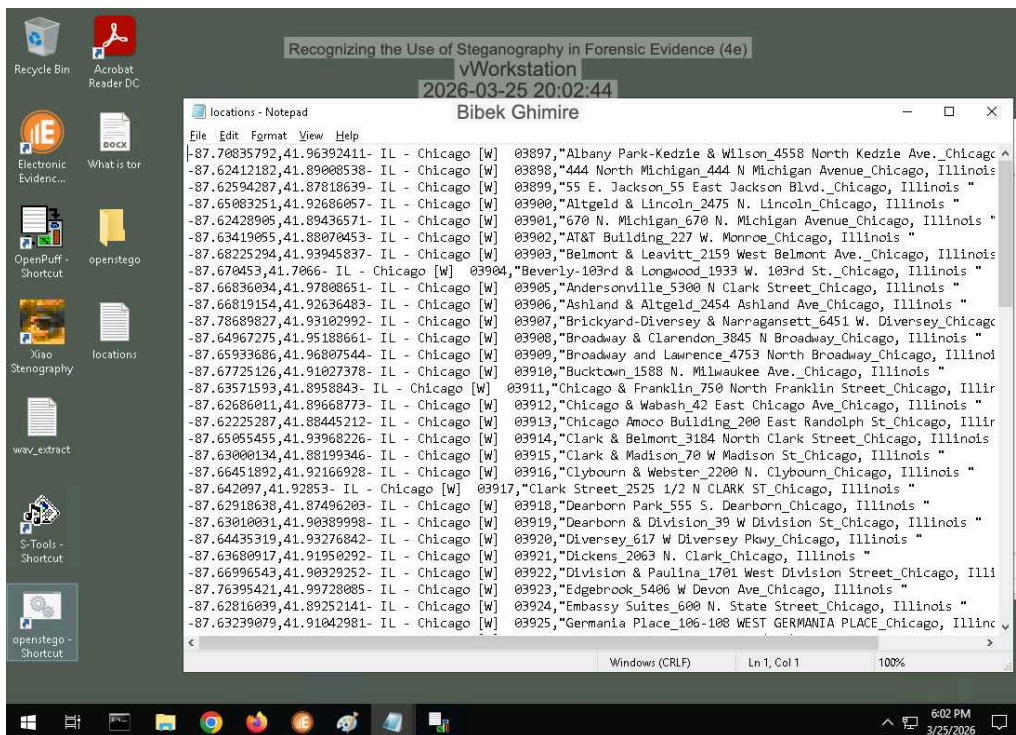


Part 3: Extract Hidden Data from Image Files

2. Record the passphrase saved in the ReadMe file.

landmarks

16. Make a screen capture showing the contents of the file extracted by OpenPuff.



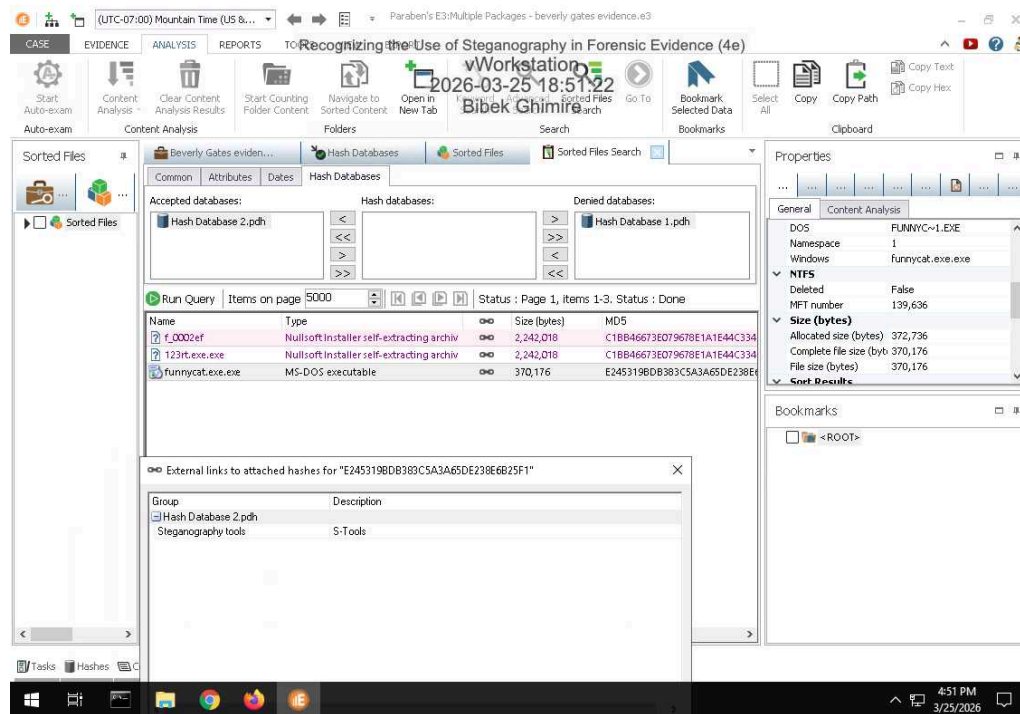
17. Describe the contents of the hidden file. How might it be relevant to the current investigation?

The use of OpenPuff to hide these coordinates suggests they are high-value locations (such as safe houses or transaction sites) that the subject intended to keep secret from investigators.

Section 2: Applied Learning

Part 1: Detect Steganography Software on a Drive Image

5. Make a screen capture showing the search result and its description.



Part 2: Detect Hidden Data in Image and Audio Files

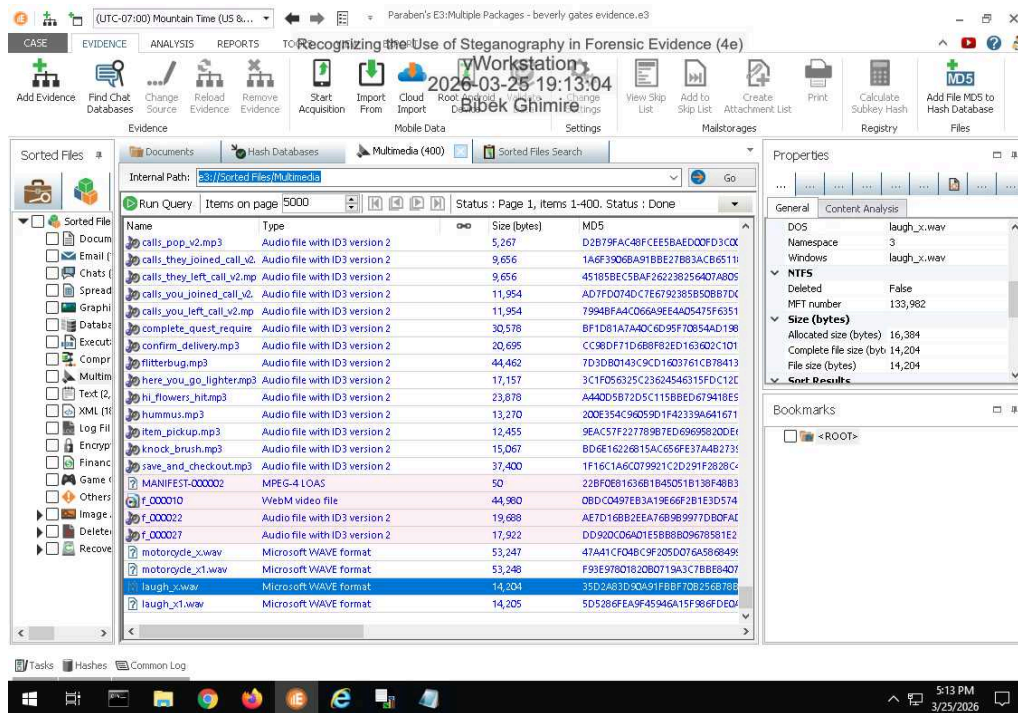
4. Identify the image file with concealed data according to the StegExpose steganalysis tool.

db9olser.gif

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7. Make a screen capture showing the WAV file sizes and hash values in E3.

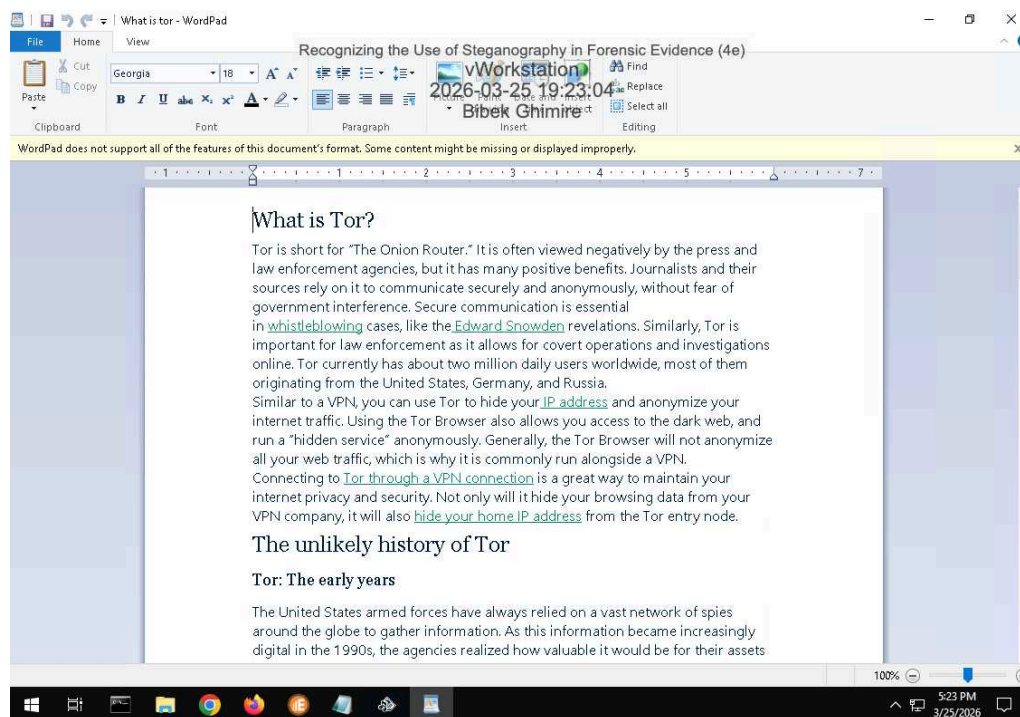


Part 3: Extract Hidden Data from Image and Audio Files

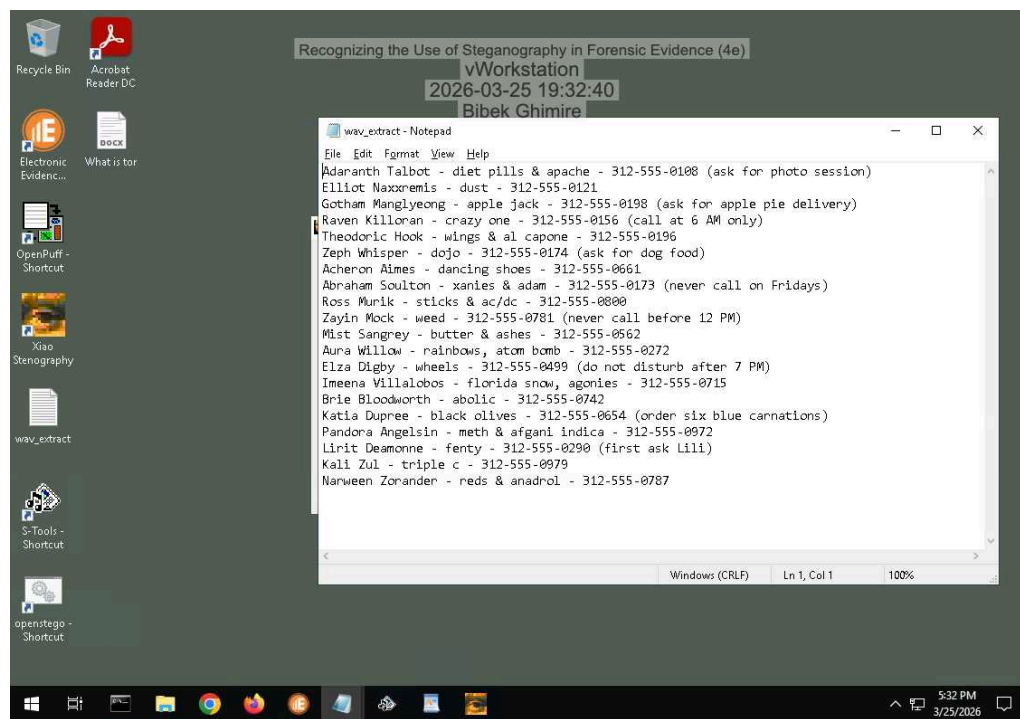
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9. Make a screen capture showing the contents of the hidden file extracted by S-Tools.



15. Make a screen capture showing the contents of the hidden file extracted by Xiao.



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16. **Describe** the contents of the two hidden files. How might they be relevant to the current investigation?

The files suggest a deliberate attempt to conduct illegal business (wav_extract) while using specialized tools to evade law enforcement (Tor). Using steganography to hide these specific files demonstrates a high level of intent to conceal evidence.

Section 3: Challenge and Analysis

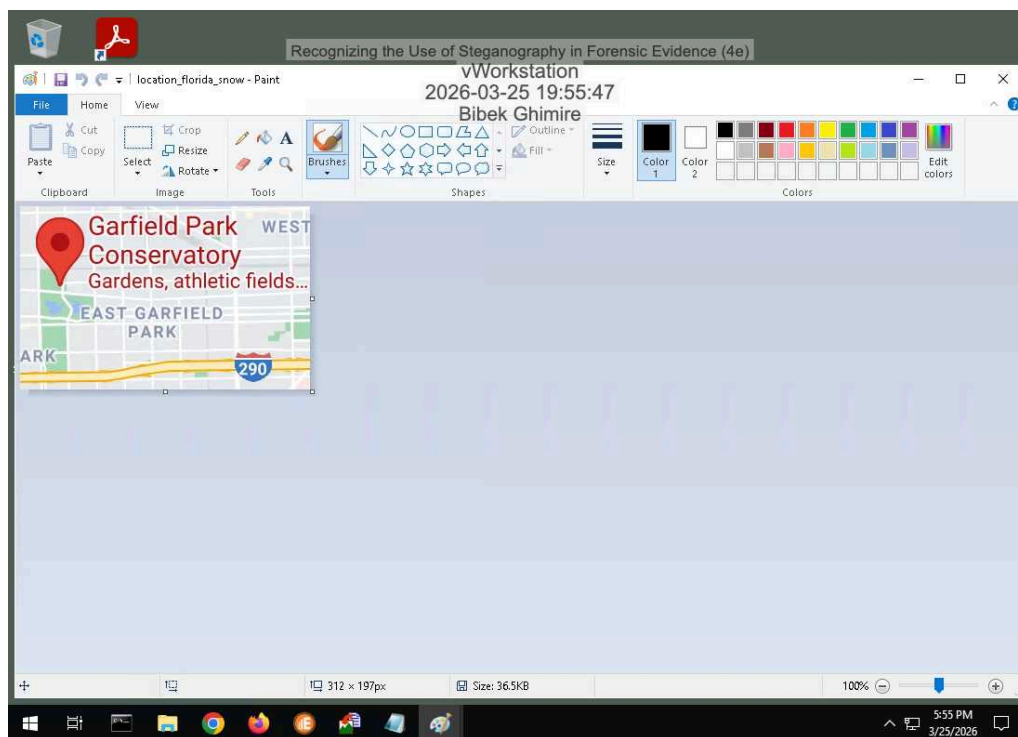
Part 1: Detect More Hidden Data

Record the names of the files that contain concealed data.

chicago.bmpchicago1.bmp

Part 2: Extract More Hidden Data

Make a screen capture showing the **first file extracted by OpenStego**.



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Make a screen capture showing the second file extracted by OpenStego.

